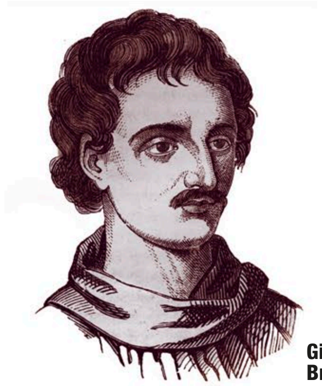
Aristarchus of Samos



Giordano Bruno

3rd Century BCE

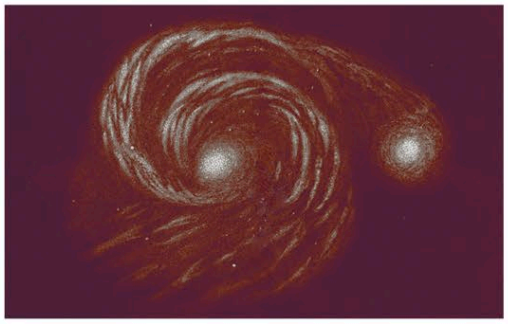
Sun-centerd Universe
The Greek astronomer Aristarchus of Samos puts forward his idea that it is the Sun that sits at the center of the Universe, with the Earth orbiting it. Aristarchus also suspects that stars are bodies similar to the Sun, but much farther away.

1543

A convincing mathematical model
Polish astronomer Nicolaus Copernicus's book *De revolutionibus orbium coelestium* is published. It contains a detailed and convincing mathematical model of the Universe in which the Sun is at the center with Earth and other planets orbiting it.

1584

An infinite multitude of stars
Italian philosopher and mathematician Giordano Bruno proposes that the Sun is a relatively insignificant star among an infinite multitude of others. He also argues that because the Universe is infinite, it has no center or specific object at its center.



Sketch of Whirlpool Galaxy

1610

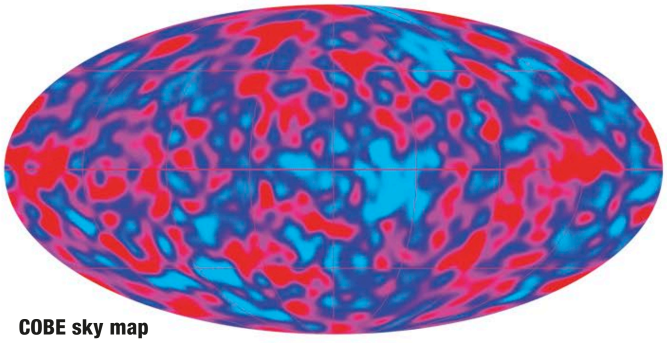
Argument against infinite Universe
German astronomer Johannes Kepler argues that any theory of a static, infinite, and eternal Universe is flawed, since in such a Universe, a star would exist in every direction and the night sky would look bright. This argument later comes to be known as Olbers' paradox.

1755

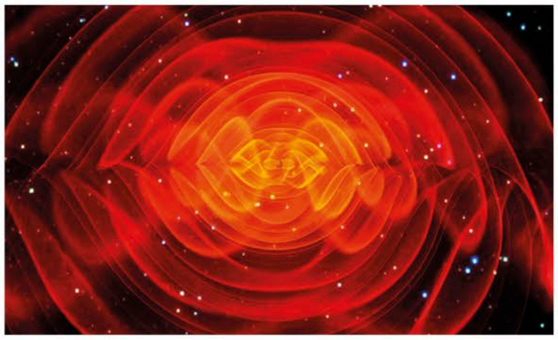
Objects exist outside our galaxy
German philosopher Immanuel Kant suggests that some fuzzy-looking objects in the night sky are galaxies outside the Milky Way Galaxy—implying that the Universe consists of more than just the Milky Way, being considerably bigger.

1905

Spacetime continuum
German physicist Albert Einstein's Special Theory of Relativity proposes that space and time form a combined continuum, spacetime. An inbuilt assumption of his theory is that no location is special—so the Universe has no center and no edge.



COBE sky map



Computer simulation of gravitational waves

1980

Inflationary Big Bang theory
The American physicist Alan Guth and colleagues suggest that the Universe expanded at a fantastically fast rate during an extremely early phase of its existence after the Big Bang. The theory helps explain the large-scale structure of the cosmos.

1992

Variations in the CMBR
Measurements by the COBE (Cosmic Backgroung Explorer) satellite reveal tiny variations in the CMBR, providing a picture of the seeds of large-scale structure when the Universe was a tiny fraction of its present size and just 380,000 years old.

1999-2001

The existence of dark energy
High-precision measurements of the CMBR and the recessional velocities of galaxies at different distances provide evidence for dark energy—a mysterious phenomenon that seems to be accelerating the Universe's expansion.

2016

Gavitational waves detected
Physicists in the US announce that they have detected gravitational waves. The existence of these waves supports the Inflationary Big Bang theory and provides further confirmation of Einstein's General Theory of Relativity.